University of Baghdad · College of Medicine

Health Anxiety, AI Chatbot Use and Cyberchondria Among Adults in Baghdad

An online cross-sectional survey

Presented by

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Supervisor

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BACKGROUND

Why this study matters

> 80%

of Iraqis are online: symptoms are searched before they reach a doctor

AI

chatbots now answer health questions conversationally and without limit

0

studies of health anxiety and cyberchondria in Baghdad before ours



Health anxiety and cyberchondria

Health anxiety

Persistent worry about having a serious illness, kept going by **misreading normal sensations** and **seeking reassurance**.

Measured with the **SHAI-14** (14 items, 0–42).

Cyberchondria

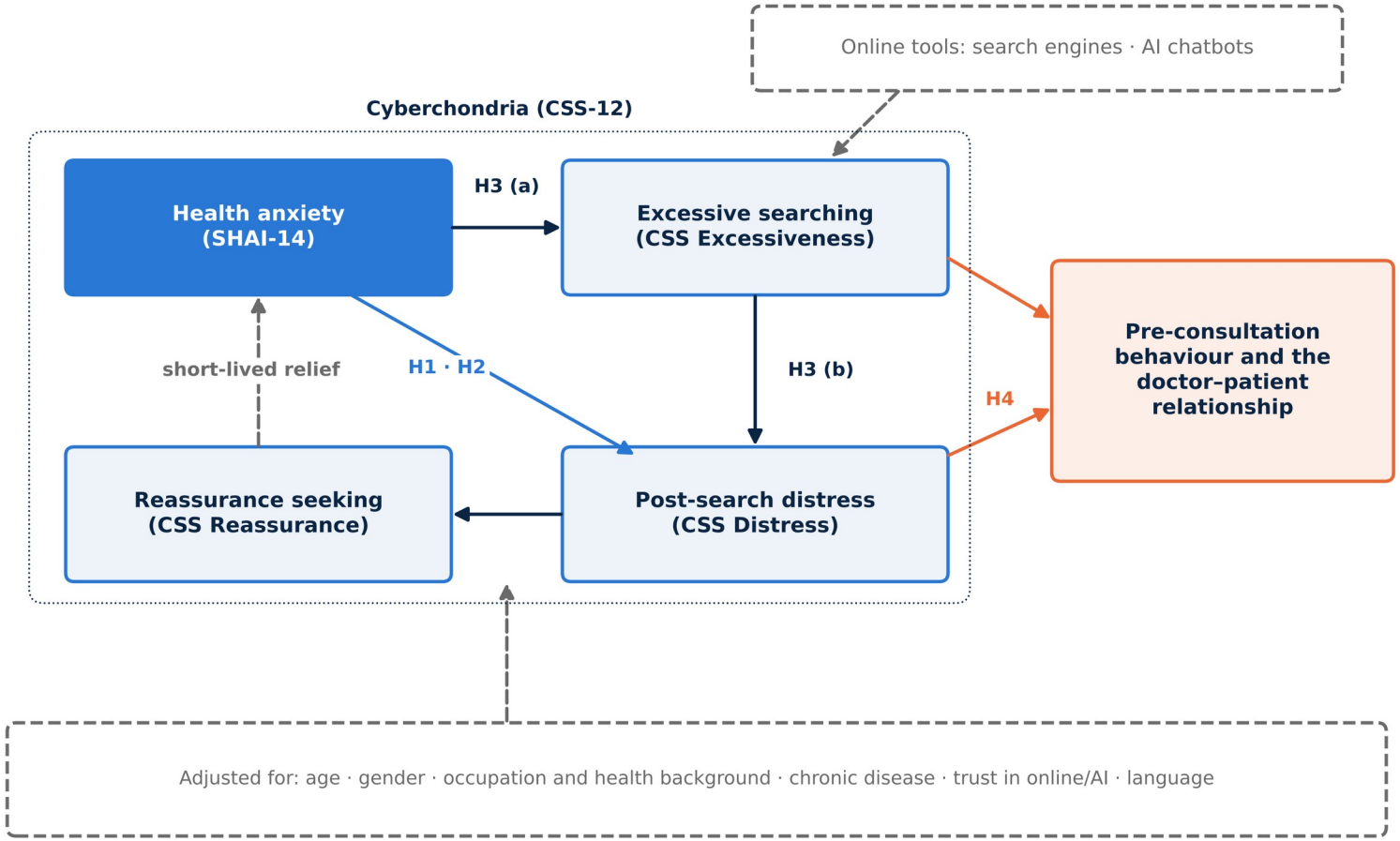
Excessive or repeated online health searching that **increases** distress instead of relieving it.

Measured with the **CSS-12** (12 items, 12–60): excessiveness · compulsion · distress · reassurance.



OUR MODEL

A vicious cycle, and four hypotheses



H1 Health anxiety ↔ cyberchondria

H2 Independent of other factors

H3 Excessive searching is a step to distress

H4 Cyberchondria ↔ care-seeking & doctor relationship

METHODS

How we did it

1

Design

Analytical cross-sectional online survey (STROBE, CHERRIES)

2

Setting

Baghdad · 17 July 2026 – 11 September 2026

3

Sampling

Convenience + snowball via Baghdad-based online groups

4

Participants

n = 300 adults ≥ 18 · consent · Arabic or English

5

Questionnaire

Bilingual Google Form · 5 parts · SHAI-14 · CSS-12

6

Ethics

Anonymous · e-consent first · [approval no.]



METHODS

Sample size and analysis

194 → 300

needed (r = 0.20, α 0.05, 80% power) → collected

±5.7%

margin of error for percentages

≈ 0.62

expected r from meta-analyses: our target is conservative

Psychometrics	Cronbach α, McDonald ω, CFA; Arabic vs English invariance
Associations	Pearson / Spearman, Mann–Whitney, Kruskal–Wallis, χ^2
Models	Hierarchical robust regression; logistic regression (AUC)
Pathways	Bootstrapped mediation (5000 resamples); moderation
Rigour	FDR correction for many tests; sensitivity analyses

02

Results

Who took part · how they search · how anxious they are · the main link · who is at risk · behaviour · AI



PARTICIPANTS

A young, educated, digitally active sample

300
participants · no missing data

62.3%
aged 20–29

56.3%
female

44.0%
medical students

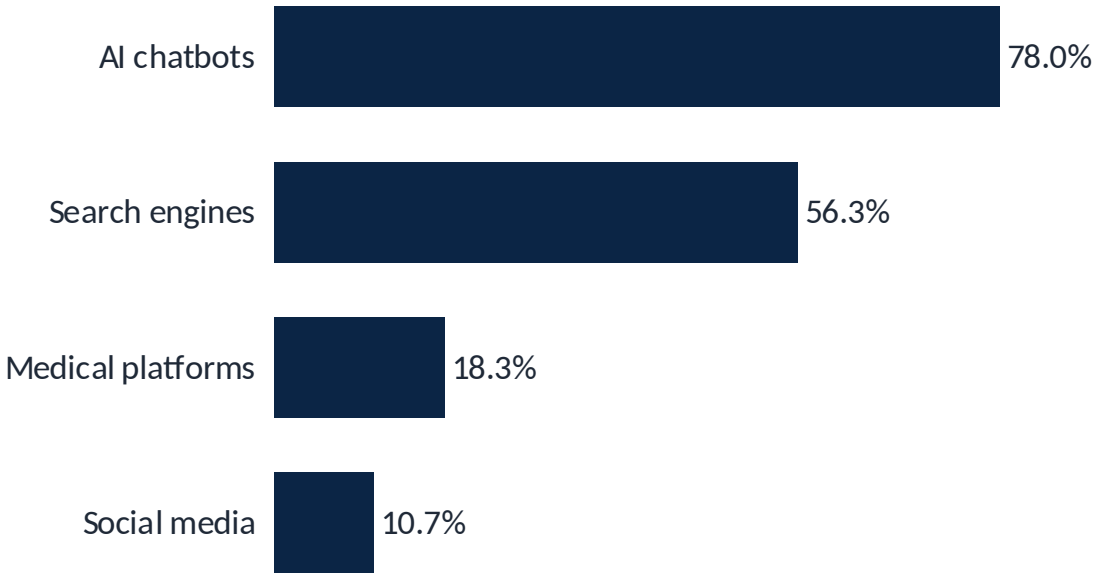
63.3%
health-related background

21.7%
chronic condition

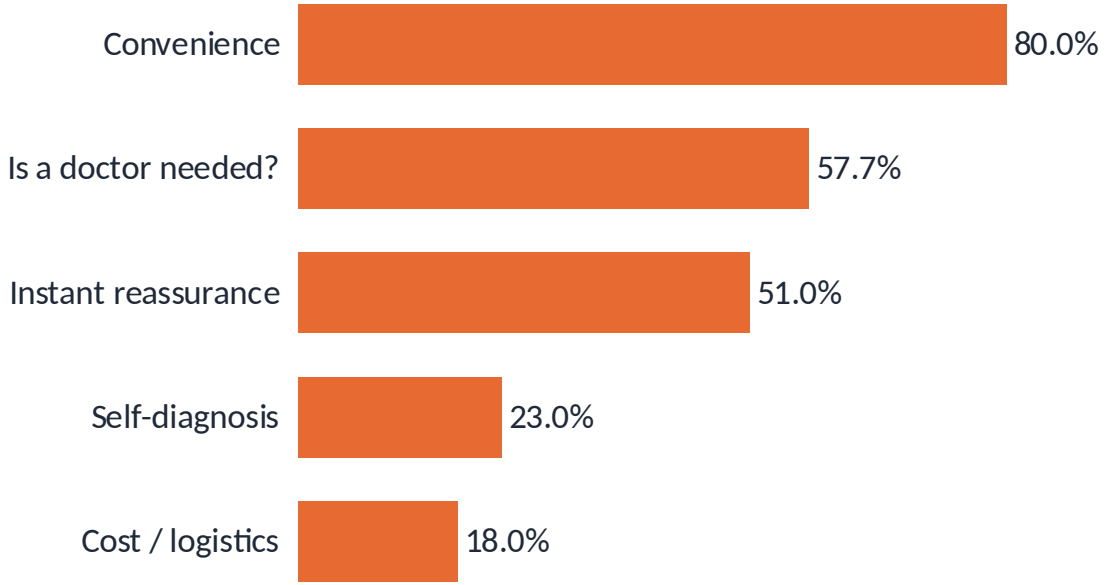
Percentages describe **this sample**; the associations, however, held in every subgroup.

AI chatbots are now the first stop

Tools used to look up symptoms (%)



Why they search (%)



88.7%
use AI incl. those who named a chatbot

41.0%
use AI without any search engine

59.3%
use ChatGPT

One in five scored in the elevated range

19%

elevated health anxiety (SHAI ≥ 18) · 95% CI 15.0–23.8%

SHAI-14 cut-off	%
≥ 15	32.7
≥ 18 (used)	**19.0**
≥ 22	7.3
≥ 29	0.7

α 0.81

SHAI-14 reliability

α 0.86

CSS-12 reliability

CFI 0.95

CSS-12 four-factor structure confirmed

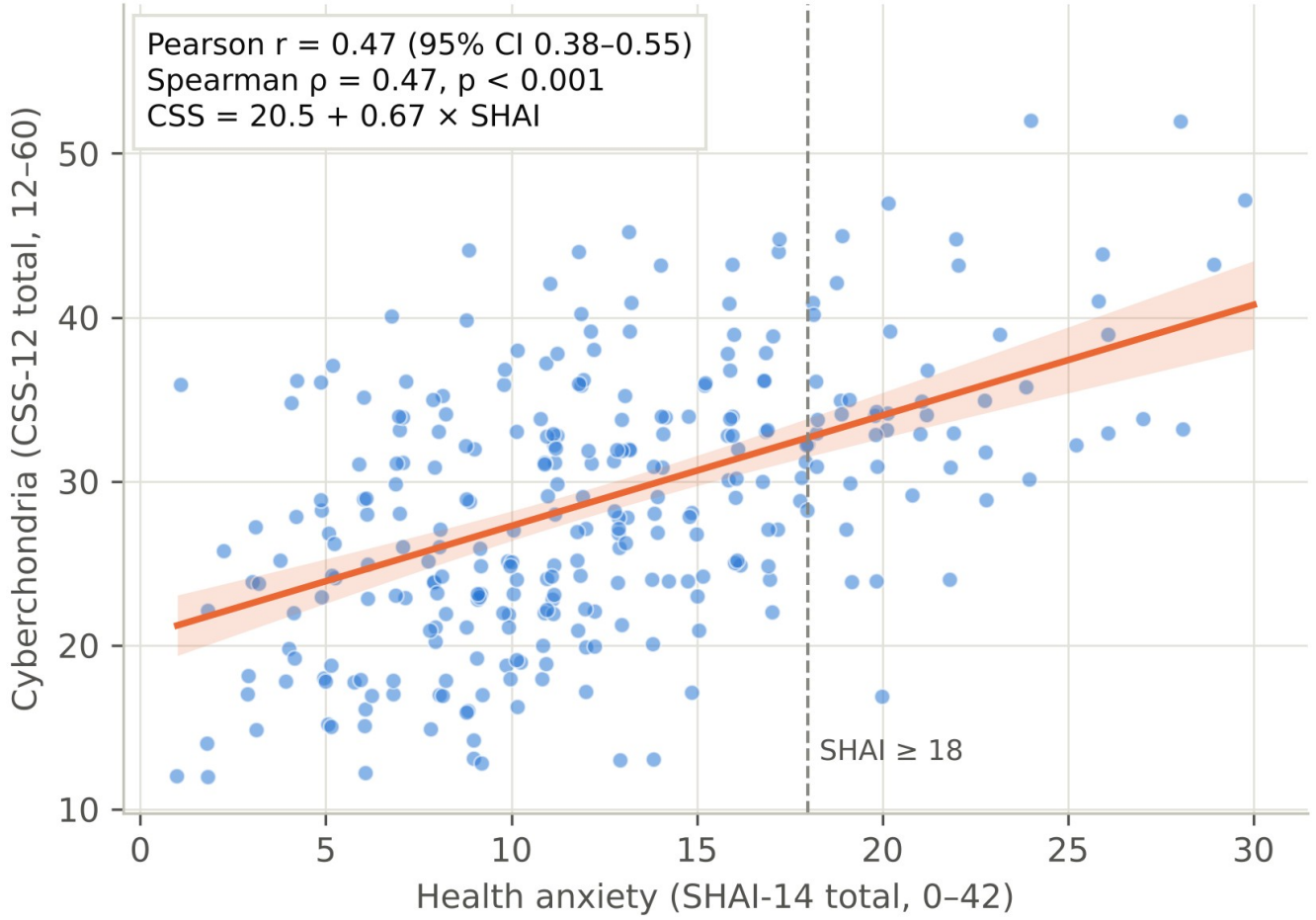
Same

CSS-12 in Arabic and English (full invariance)

SHAI mean 12.3 / 42 · CSS mean 28.9 / 60 · excessiveness highest, compulsion lowest

MAIN RESULT

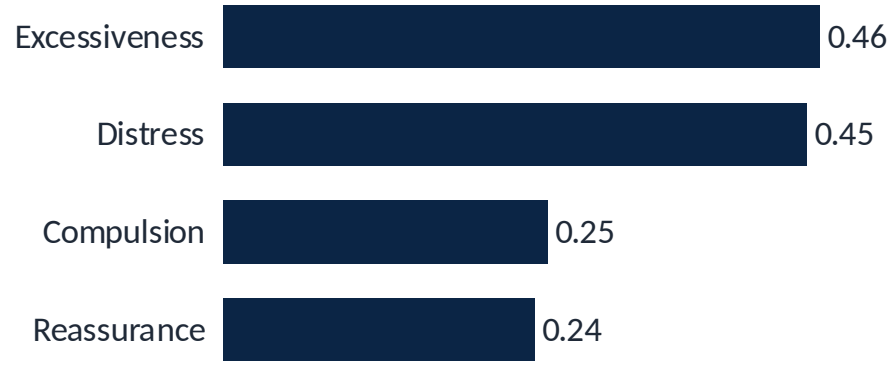
Health anxiety and cyberchondria rise together



r = 0.47

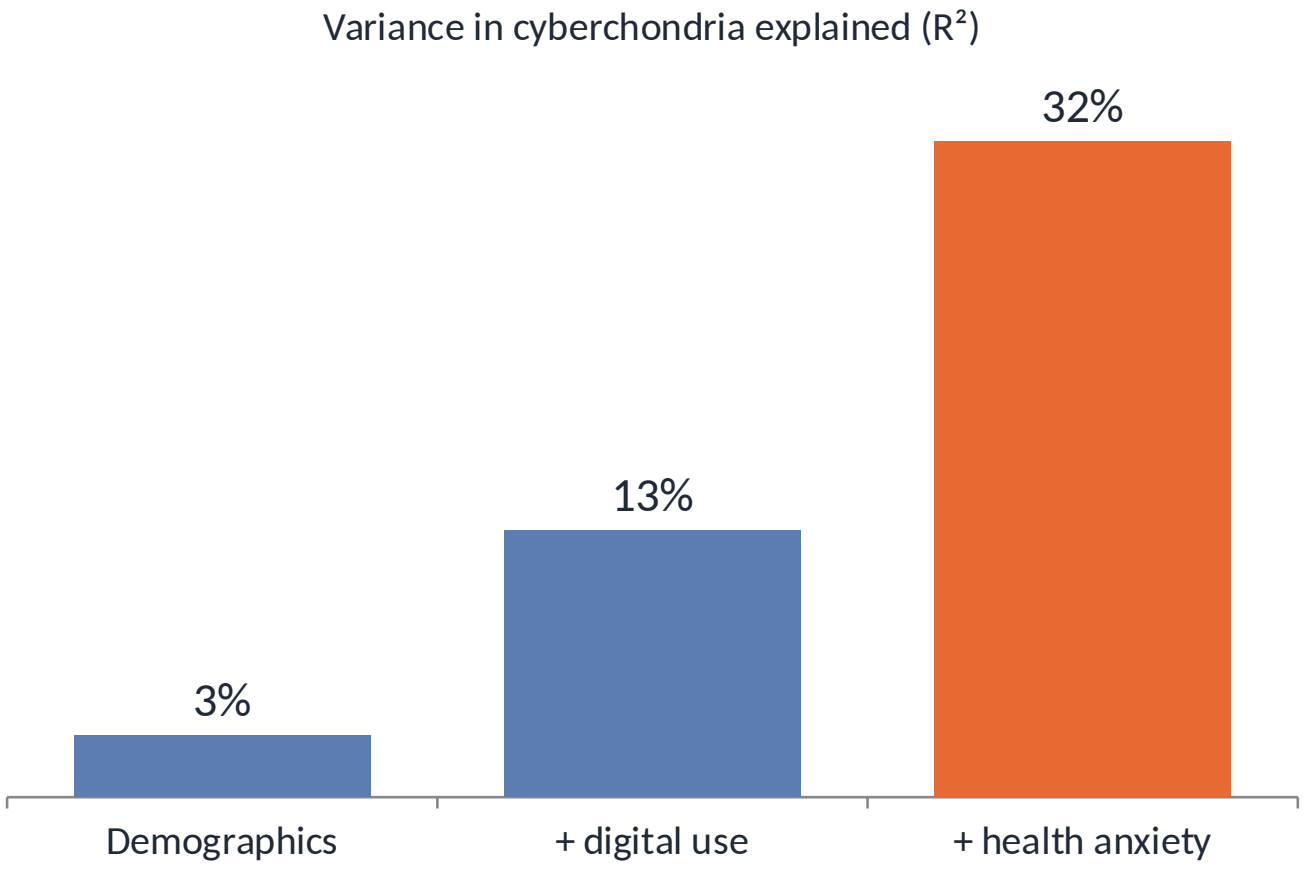
95% CI 0.38–0.55 · p < 0.001 · partial r = 0.48

Correlation with each facet



H1 and H2 supported

How anxious you are matters more than who you are



+18.4%

extra variance explained by health anxiety

β = 0.48

strongest predictor · B = 0.68 CSS points per SHAI point

Not robust

AI use: +5.3 (inclusive) but +1.5, p = 0.31 (strict)

Fear and interpretation, not noticing

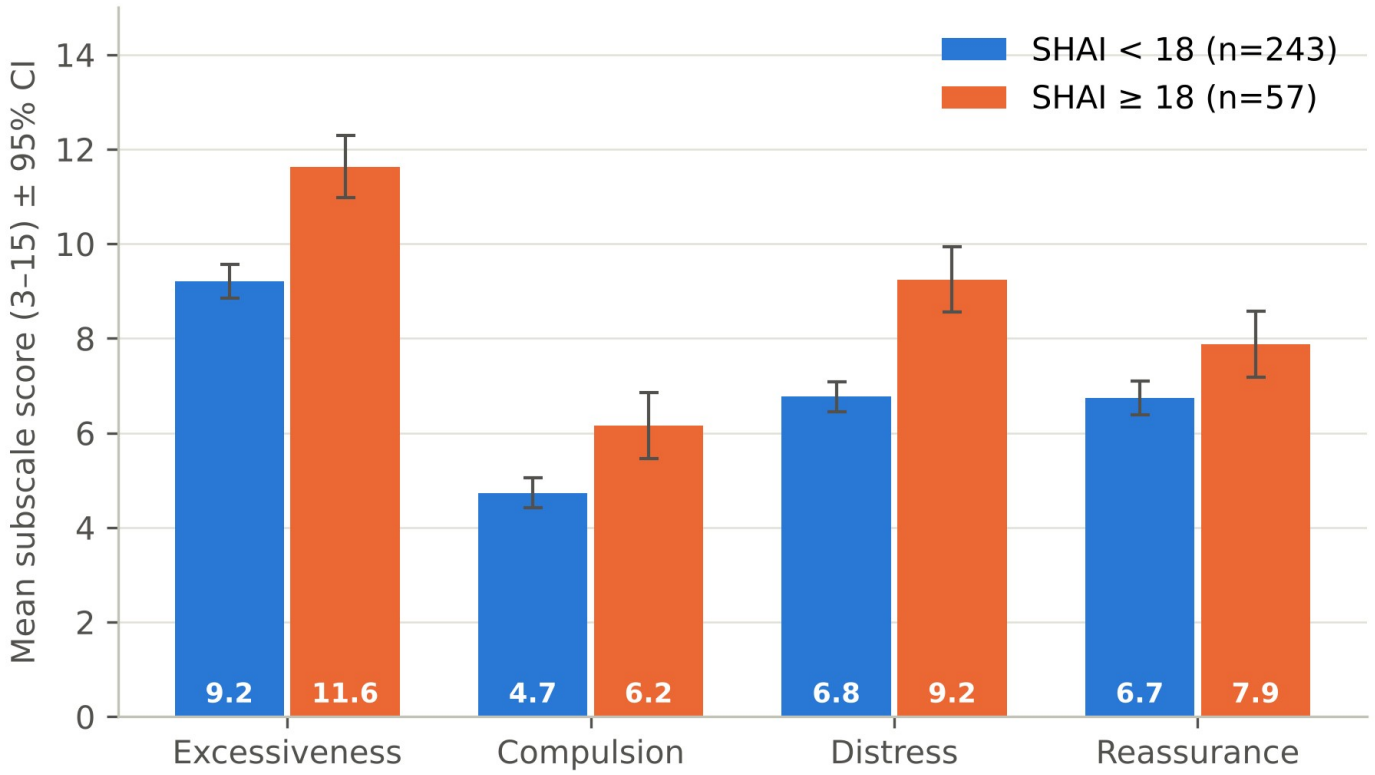


Figure 10: CSS facets by health-anxiety status (mean ± 95% CI)

ρ 0.43

Fear of serious illness ↔ panic on reading of a rare condition

ρ 0.42

Interpreting sensations ↔ searching every sensation

≈ 0

Merely noticing aches or sensations

111 of 168 item pairs significant after FDR

Young people who search for reassurance

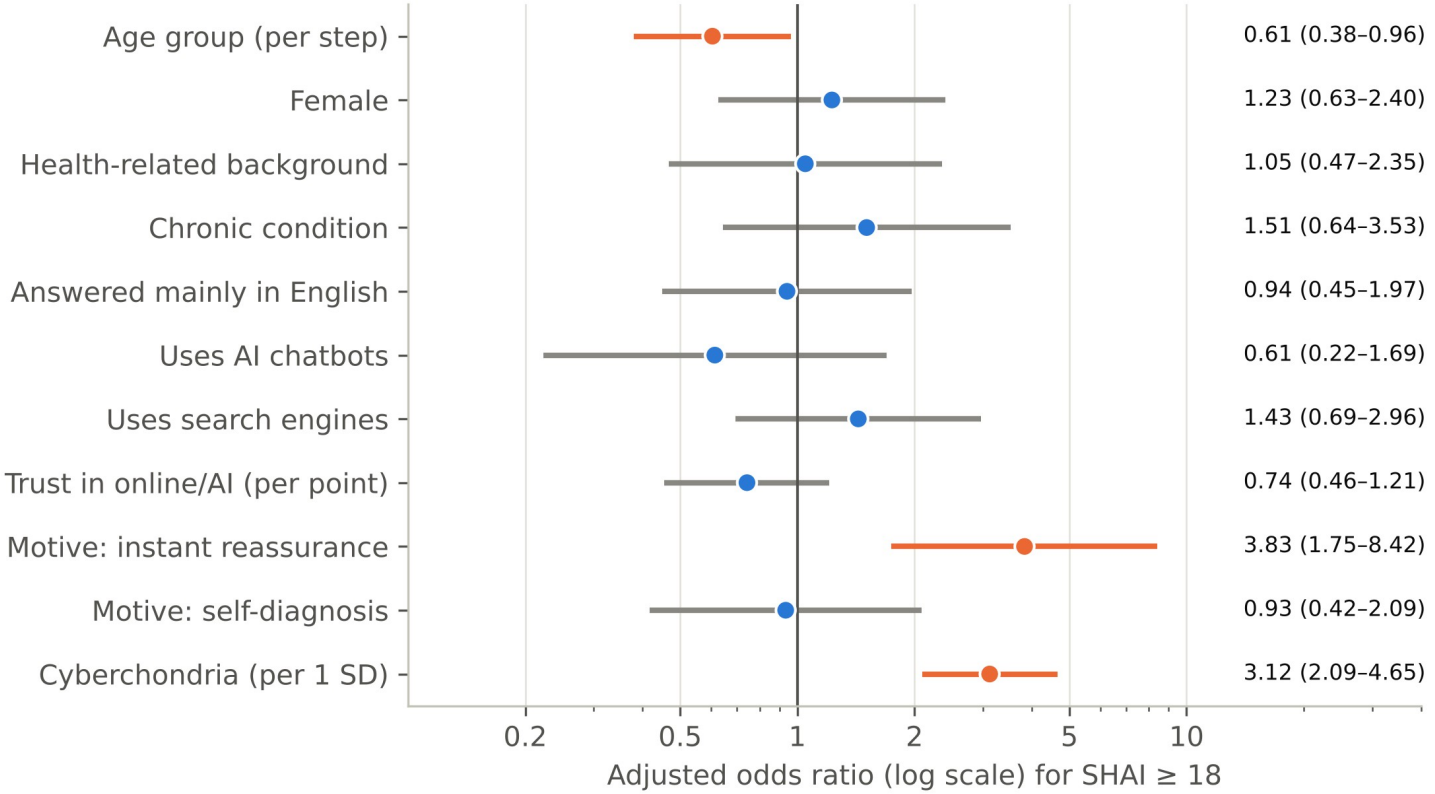


Figure 8: adjusted odds ratios; orange = p < 0.05

3.12×
odds per SD of cyberchondria

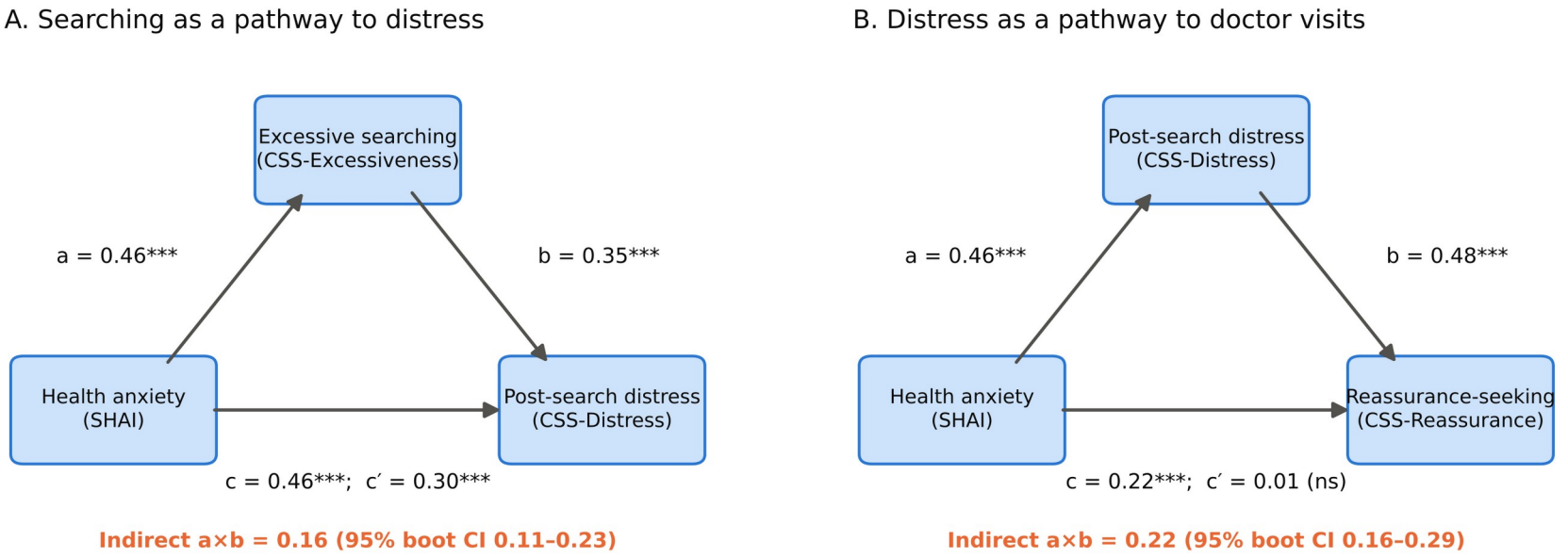
3.83×
if searching for instant reassurance

0.61×
per older age group (protective)

AUC 0.79
after optimism correction (apparent 0.83)

PATHWAYS

Compatible with a cycle, not proof of one



Standardised coefficients adjusted for age, gender, health background, chronic illness, trust and AI use; 5,000 bootstrap resamples. * p<0.05, ** p<0.01, *** p<0.001.

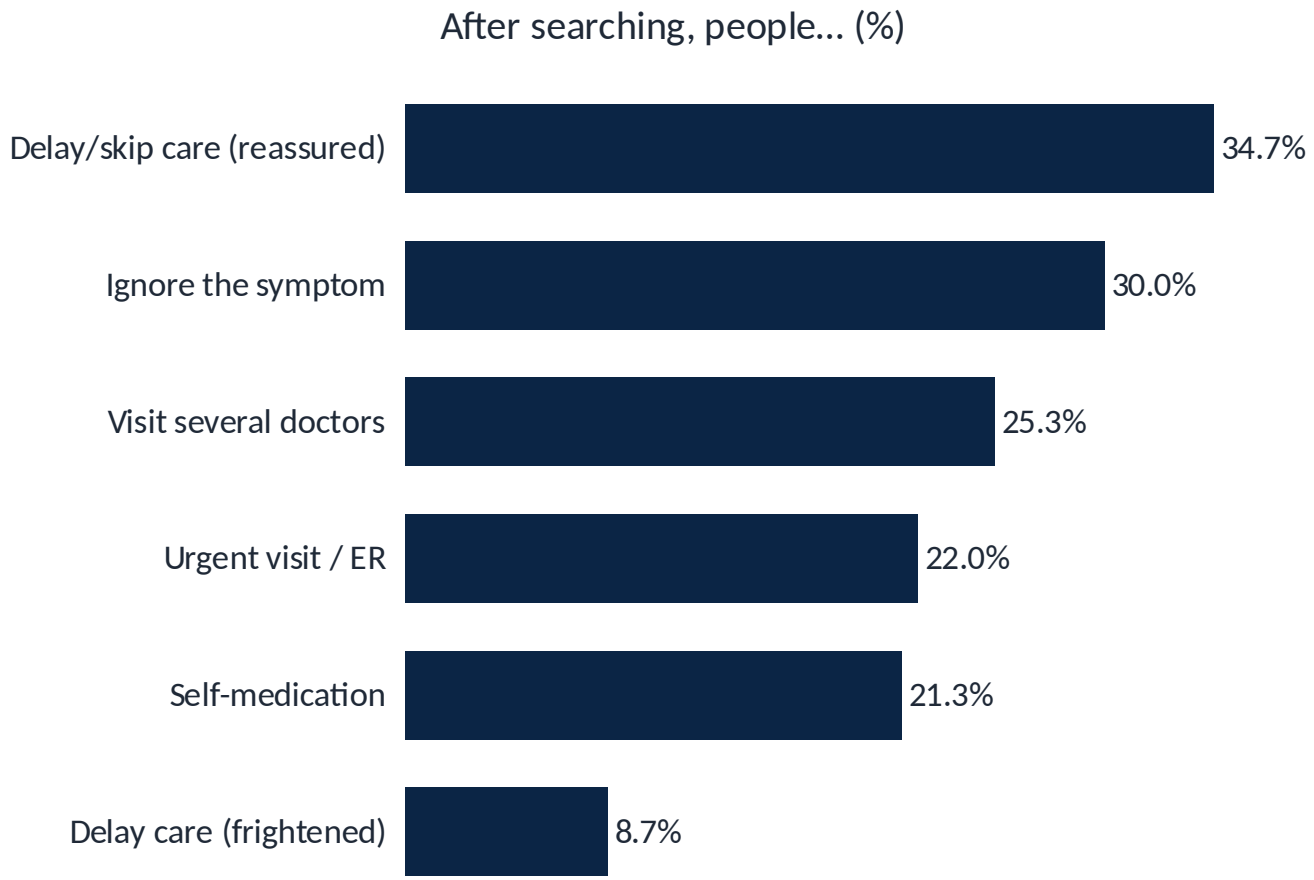
Figure 9: mediation models (standardised; 5000 bootstrap resamples)

0.16
indirect via excessive searching (CI 0.11–0.23)

0.01
direct effect left after distress (not significant)

0.15
reverse direction also significant

Searching shapes the consultation



31.7%
never tell their doctor (fear of judgement)

34.7%
say searching strained their relationship with doctors

OR 0.57
higher cyberchondria → less hiding (q = 0.012, robust)

AI CHATBOTS

No reliable independent effect of AI

+5.3

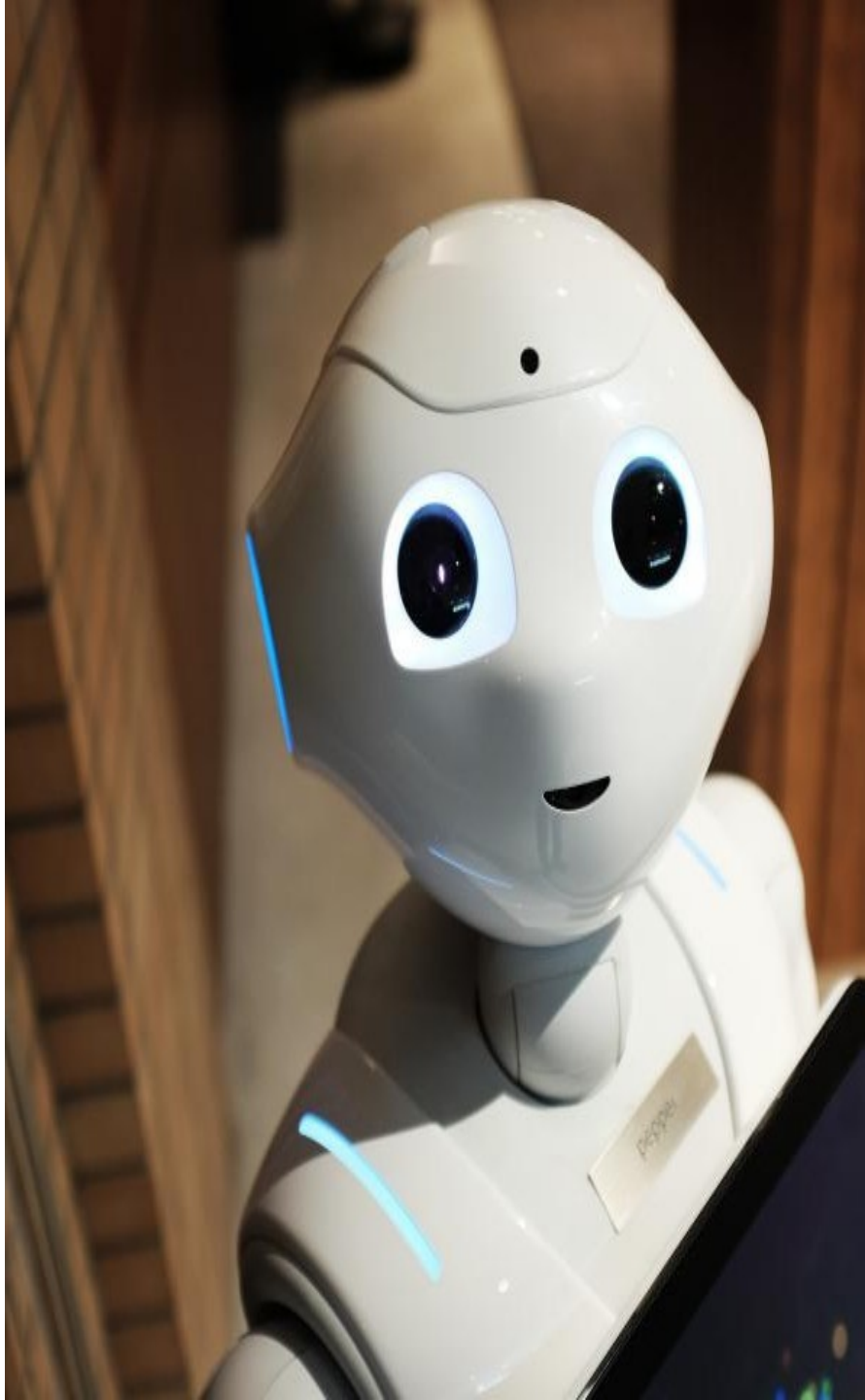
CSS points · inclusive definition

+1.5

strict definition · p = 0.31

1. 32 respondents named a chatbot but did not tick “AI”; they scored highest
2. Only 34 true non-users: a small comparison group
3. AI use did not change the health anxiety ↔ cyberchondria link

Absence of evidence, not evidence of absence.



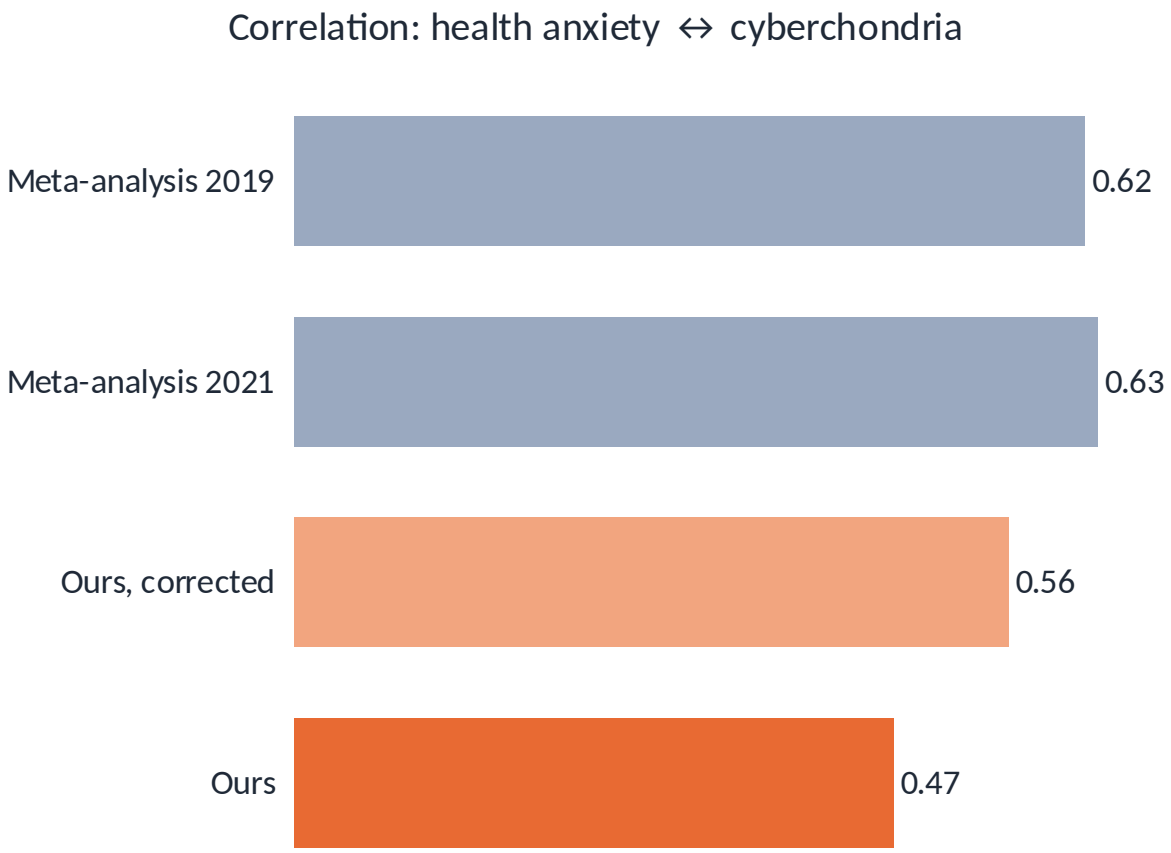
03

Discussion

How we compare · why · AI and the consultation · medical students · limitations · recommendations



Same direction, weaker link: four reasons



1 **Measurement error**

corrected for reliability, r ≈ 0.56

2 **Narrow range**

low health anxiety (mean 12.3) weakens r

3 **Who answered**

online recruitment favours confident users

4 **Short form**

most studies used the 33-item CSS

COMPARED WITH THE ARAB REGION

Consistent with regional studies



Saudi Arabia · Jeddah

Common; linked with smartphone addiction

Saudi Arabia · primary care

Linked with health anxiety

Lebanon

Arabic CSS-12: same four-factor structure

Egypt · 1435 medical students

85.8% moderate cyberchondria

AI users and medical students

AI chatbots

Our use **78.0%–88.7%** vs ≈ 20% (US 2024) and 21% (2025).

Young sample: **96%** at 18–19 y vs **72%** at ≥ 40 y.

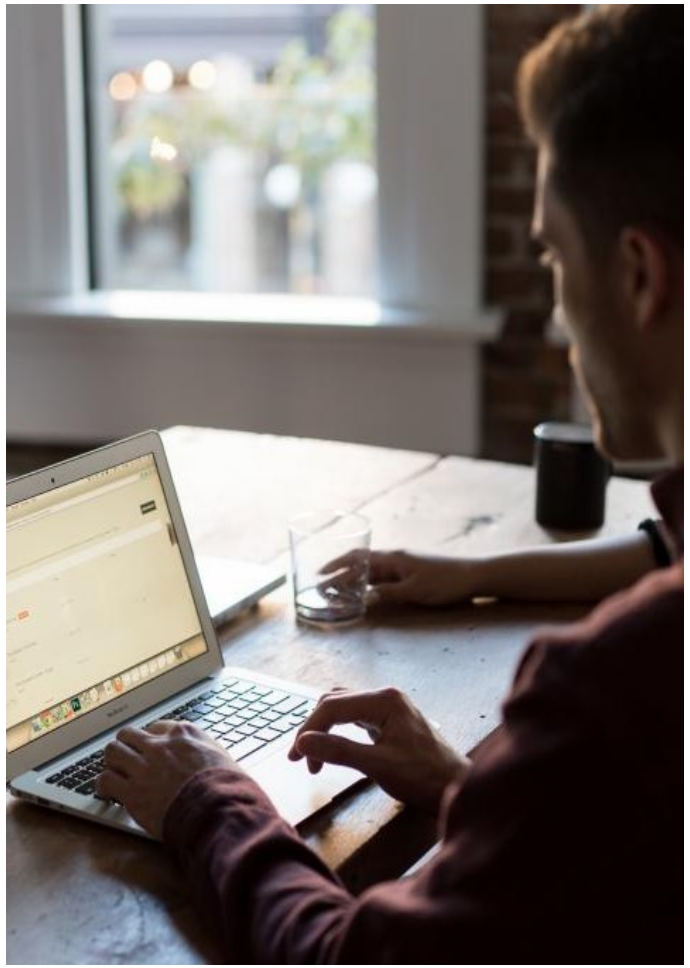
Conversational AI may invite “one more question”.

Medical students

44.0% of our sample.

Higher health anxiety **disappeared after adjustment** (younger age).

Same cyberchondria as others: **knowledge did not protect**.



WHAT IT MEANS FOR THE CONSULTATION

Ask every patient what they searched

34.7%
delay care when reassured: saves visits, but risks delayed diagnosis

31.7%
never tell the doctor: hidden information in the consultation

H4
only partially supported: disclosure was the robust finding

Clinical message: ask, without judgement, what patients searched or asked an AI.



LIMITATIONS

What limits our conclusions, and what we did

1	**Cross-sectional design**	No causal direction	→ Reverse models tested; “compatible with”
2	**Convenience sample**	Young, 44% medical students	→ Percentages not generalised; associations consistent
3	**Arabic wording**	No back-translation	→ Equal reliability; CSS-12 fully invariant
4	**Single-item measures**	Behaviour, disclosure, trust	→ Reported descriptively; labelled exploratory
5	**Many tests**	Risk of false positives	→ FDR correction; robust vs exploratory

RECOMMENDATIONS

From findings to action

Clinicians

Ask non-judgementally; treat repeated searching as a sign of health anxiety; teach AI literacy

Ministry of Health

May consider Arabic digital & AI health-literacy campaigns and reliable triage sources

AI developers

May consider anxiety-aware answers that do not escalate rare diagnoses

Researchers

Longitudinal studies; probability samples; trials: only 9 registered, none in Iraq

CONCLUSION

Three things to remember

1

AI chatbots are the first stop

78.0% ticked them (88.7% incl. named), more than search engines.

2

Health anxiety ↔ cyberchondria, r = 0.47

Robust, compatible with a cycle; no reliable independent AI effect.

3

Searching shapes the consultation

34.7% delay care when reassured; 31.7% never tell the doctor.

Ask every patient, without judgement, what they searched or asked an AI.

r = 0.47

Health anxiety and cyberchondria travel together.

Thank you · questions are welcome

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